

NetSight: AI-Assisted Network Controller for Amarisoft 5GC

The objective of this work is to design and implement an AI-assisted control framework for the Amarisoft 5G Core that enables network management through natural language interaction. The framework combines a locally deployed Large Language Model (LLM) with a controller that executes management actions on the Amarisoft 5GC. Prompt engineering and network-specific contextual information are used to guide the LLM in interpreting operator requests and generating structured outputs, which are validated and translated into executable actions by the controller. The framework will be evaluated through the implementation of representative network management use cases to assess its capabilities, identify implementation challenges, and discuss its practical limitations.

Use Case 1: Basic Configuration Management

The first use case focuses on performing basic configuration changes on the Amarisoft 5GC using natural language commands. The LLM shall interpret an operator request and generate a structured response that enables the controller to identify the required configuration parameters, apply the requested modifications, and verify that they have been successfully executed.

The implementation will initially focus on a small set of low-risk configuration changes to establish the end-to-end workflow. Examples include modifying QoS parameters (e.g., 5QI, GBR, or ARP settings), adjusting network timer values (e.g., T3412 or transaction timeout parameters), and enabling or disabling selected network functions or configuration options.

The primary objective of this use case is to demonstrate the reliable translation of simple natural language requests into valid and executable configuration changes.

Use Case 2: Multi-component Configuration Management

The second use case extends the framework to network management tasks that require coordinated modifications across multiple configuration entities. Unlike the first use case, which focuses on isolated parameter changes, this use case evaluates whether the LLM can identify all configuration elements affected by a higher-level operator request and generate a consistent set of modifications for the controller. Examples include:

- configuring connectivity between the AMF and an external network function (e.g., UDM, AUSF, or EIR)
- configuring a new network slice or DNN together with its associated QoS parameters.

This use case is used to evaluate the LLM's ability to identify dependencies between multiple configuration entities and generate consistent modifications required to satisfy a higher-level operator request.

Use Case 3: Network Diagnosis and Corrective Actions

The third use case focuses on troubleshooting network issues using the LLM as a decision support tool. The LLM shall analyse operator-provided information, such as Amarisoft log files, configuration files, or Remote API outputs, to identify likely causes of a network issue and recommend appropriate corrective actions. Following operator verification, the controller may execute the approved corrective action on the Amarisoft 5GC.

Examples include identifying the cause of failed UE registration attempts, detecting inconsistencies between related configuration files or network functions, and recommending corrective actions such as updating configuration parameters or re-establishing connectivity to external network functions.

This use case aims to evaluate the LLM's ability to reason over network information and generate reliable recommendations for troubleshooting and corrective actions.